Manuel Rodriguez Berdud

Lakeland, FL (Open to Relocate) | manuelrguezberdud@outlook.com | (402) 504-8262 | LinkedIn: manurguez03

Data Science graduate student with a **strong quantitative background**, interested about using **statistics, ML, and AI** to solve real-world problems, especially in **sports analytics**, where technology meets my love for sports.

Education

M.S. Data Science, FLORIDA POLYTECHNIC UNIVERSITY Aug 2025 – May 2027
GPA: 3.89/4.0 | SHPE Member | **Related Coursework**: Time Series Analysis, Econometrics, Optimization, Machine Learning, Data Wrangling & EDA, Computational Data Analysis

B.S. Mathematics and Physics, CONCORDIA UNIVERSITY NEBRASKA Aug 2021 – May 2025
GPA: 3.89/4.0 | Academic All-American (2024, 2025) | **NAIA Athlete**: Tennis & Soccer Scholarships

Projects

Causal Impact of Lucky Loser Entry on Tennis Careers – Master's Thesis Apr 2026 – Present
- Built a longitudinal dataset of **3,000+ player-month observations** across **140+ tennis players** and 16 years of records, using Python and LLM-assisted pipelines to integrate ATP rankings and match results.
- Conducted hypothesis testing with **95% confidence intervals** to compare Grand Slam Lucky Loser selection probabilities against theoretical random-draw expectations.
- Designed a **quasi-experimental framework using Difference-in-Differences (DiD)** to estimate the causal impact of Lucky Loser selection on players' long-term career outcomes.

Biomechanics-Based Pitch Command Analysis – Lakeland Flying Tigers Aug 2026 – Present
- Processed **300,000+ pitch records** and 200+ variables into an analysis-ready subset of **81,000+ pitches** and 100+ biomechanical predictors.
- Applied **Random Forest-based Boruta feature selection** and cross-validation to identify key pitch-command predictors; validating a reduced feature set for **predictive modeling**.

LLM-Powered Data Wrangling and EDA – NYC Marathon Feb 2026 – Apr 2026
- Built an **automated LLM-powered workflow** to clean, transform, and analyze **55,000+ marathon records** using R
- Used exploratory analysis and **Tableau** dashboards to visualize runner performance patterns and trends.

Experience

Graduate Research Assistant, Florida Polytechnic University – Lakeland, FL Aug 2025 – Present
- Worked with faculty across **3+ research projects**, reviewing statistical methodologies, evaluating analytical approaches, and refining research findings to strengthen academic manuscripts.
- Developed **Python-based data collection** and processing pipelines, transforming raw, unstructured data into analysis-ready datasets for **statistical modeling and predictive analysis**.

Tech Consulting Intern, Structurit – Madrid, Spain May 2026 – Jul 2026
- Helped with **5+ technical proposals**, translating client requirements into **AWS cloud architectures**, data integration strategies, and implementation roadmaps for **Industry 4.0 initiatives**.
- Contributed to technical solution planning involving **NeuraPlant** (industrial IT/OT data integration) and **DocuTons** (private generative AI), evaluating cloud services, security requirements, and deployment strategies.
- Completed **AWS Certified AI Practitioner training**, gaining foundational knowledge of AWS AI/ML services, generative AI, and responsible AI practices.

Technical Skills

Programming & Data: Python, R, SQL, pandas, NumPy, tidyverse, API Testing

Statistics & Machine Learning: Regression, Classification, Clustering, Neural Networks, Time Series, Causal Inference, Feature Selection Techniques

Visualization & Tools: Tableau, Power BI, Matplotlib, Jupyter Notebook, Git/GitHub, Excel

AI & Automation: LLM Integration, Prompt Engineering, AI-Assisted Data Pipelines, Agentic AI, Generative AI

Languages: Spanish (Native), English (Fluent), Portuguese (Intermediate)

Leadership

Dining Service Supervisor, Chartwells – Seward, NE Oct 2021 – May 2025
- Supervised and coordinated a team of **6+ employees** during daily dining operations

College Tennis Player, Concorida University Nebraska – Seward, NE Aug 2021 – May 2025
- Developed strong **teamwork**, **discipline**, **time management**, and **resilience** while balancing with academics

Tennis Coach, Weils Tennis Academy – Ojai, CA Jun 2025 – Aug 2025
- Coached and mentored **15+ young players**, improving their technical skills and competitive performance